Gaussian Maxima and the Weak Simplex Conjecture

Abhijeet Mulgund
University of Illinois Chicago

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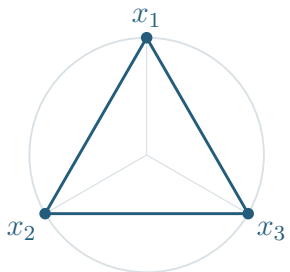






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The Weak Simplex Conjecture



$d = 2, \quad m = 3$

$m = d + 1 \geq 2$ equiprobable signals

$x_i \in \mathbb{R}^d, \qquad \|x_i\| = 1.$

$Y = \lambda x_K + Z, \quad Z \sim N(0, I_d).$

K uniform; Z independent of K; $\lambda > 0$.

Which configuration maximizes the average probability of correct ML decoding?

Main result

The regular simplex is the unique optimum for every $\lambda > 0$.

Reduction to Gaussian maxima

$$G = \Big(\langle x_i, x_j \rangle\Big)_{i,j}, \qquad \xi \sim N(0, G).$$

Exact correct-decoding probability

$$P_{\mathrm{c}}(x_1, \ldots, x_m; \lambda) = \frac{e^{-\lambda^2/2}}{m}\, \mathbb{E} \exp(\lambda \max_{1 \le i \le m} \xi_i)\,.$$

The codebook enters only through its Gram matrix G.

Common-noise transformation

$\alpha = (m-1)/m$, $\quad J = \mathbf{1}\mathbf{1}^\top$, $\quad Z_0 \sim N(0,1)$ independent of ξ.

$$X = \sqrt{\alpha}\,\xi + \frac{Z_0}{\sqrt{m}}\mathbf{1},$$

$$R = \mathrm{Cov}(X) = \alpha G + J/m.$$

Every codebook gives a correlation matrix R with

$$R - \frac{1}{m}J = \alpha G \succeq 0.$$

Regular simplex **Independent Gaussians**

$$G_\triangle = \frac{m}{m-1}\left(I_m - \frac{1}{m}J\right) \xrightarrow{\quad \alpha G_\triangle + J/m \quad} I_m$$

Gaussian stochastic domination

Theorem Let R be an $m \times m$ correlation matrix with $R - J/m \succeq 0$.

$$X \sim N(0, R), \qquad Z_1, \ldots, Z_m \overset{\text{iid}}{\sim} N(0,1).$$

For every real common threshold c,

$$\mathbb{P}\{\max_i X_i \le c\} \;\ge\; \Phi(c)^m \;=\; \mathbb{P}\{\max_i Z_i \le c\}.$$

Equivalently, $\max_i X_i \le_{\text{st}} \max_i Z_i$.

If $R \neq I_m$, then $\mathbb{P}\{\max_i X_i \le c\} > \Phi(c)^m$ for all $c \in \mathbb{R}$.

Independently: Su–Yang–Xu–I (2026)

Proof idea: adaptive centering

Centered product inequality Milman–Nakamura–Tsuji (2026)

Bounded log-concave $f_i : \mathbb{R} \to [0, \infty)$, with $\int f_i \, d\gamma_1 > 0$:

$$\int z f_i(z) \, d\gamma_1(z) = 0 \ (\forall i) \quad \Longrightarrow \quad \mathbb{E} \prod_i f_i(X_i) \geq \prod_i \int f_i \, d\gamma_1.$$

Half-line indicator

$$f(z) = \mathbf{1}_{\{z \leq c\}}.$$

$$\int z f(z) \, d\gamma_1(z) = -\varphi(c) \neq 0.$$

Centered tilted factor

$$r(s) = \varphi(s)/\Phi(s), \quad H(s) = s + r(s).$$

$$f_s(z) = e^{r(s)z} \mathbf{1}_{\{z \leq H(s)\}}.$$

$$\int z f_s(z) \, d\gamma_1(z) = 0.$$

Proof idea: alignment and comparison

Fix $c \in \mathbb{R}$. Write $a_i = r(s_i)$ and $b_i = s_i + a_i$.

We need a common threshold after the Gaussian shift

$$b - Ra = c\mathbf{1}, \qquad \text{equivalently } s + a - Ra = c\mathbf{1}.$$

We also need the inequality

$$\mathbb{P}\{X \le c\mathbf{1}\} \;\ge\; \exp\Big(\tfrac{1}{2}a^\top (I_m - R)a\Big) \prod\nolimits_i \Phi(s_i) \;\ge\; \Phi(c)^m.$$

Under $R - J/m \succeq 0$, a variational maximizer gives both:

Stationarity	aligns the thresholds.
Value comparison	gives the second inequality.

Consequences

Weak Simplex Conjecture

$m = d + 1$ equiprobable, equal-energy signals in $\mathbb{R}^d$.
The regular simplex is uniquely optimal for every $\lambda > 0$.

Simplex Mean Width Conjecture

For $m \geq 2$, choose $G \succeq 0$ with $G_{ii} = 1$ to maximize

$$\mathbb{E} \max_{1 \leq i \leq m} \xi_i, \qquad \xi \sim N(0, G).$$

Unique maximizing covariance: $G_{\triangle}$.

Exact finite-energy AWGN optimum

Uniform prior, no feedback;
per-codeword energy constraint, unrestricted uses.



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